

IMPROVING BAGGING I

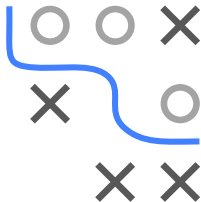
How to make $\mathbb{E}_{xy} [\Delta (f^{[M]}(\mathbf{x}))]$ as large as possible?

$$\mathbb{E}_{xy} [L(y, f^{[M]}(\mathbf{x}))] = \frac{1}{M} \sum_m \mathbb{E}_{xy} [L(y, b^{[m]})] - \mathbb{E}_{xy} [\Delta (f^{[M]}(\mathbf{x}))]$$

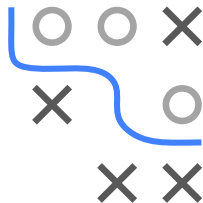
Assume $\mathbb{E}_{xy} [b^{[m]}] = 0$ for simplicity, $\text{Var}_{xy} [b^{[m]}] = \mathbb{E}_{xy} [(b^{[m]})^2] = \sigma^2$,
 $\text{Corr}_{xy} [b^{[m]}, b^{[m']}] = \rho$ for all m, m' .

$$\Rightarrow \text{Var}_{xy} [f^{[M]}(\mathbf{x})] = \frac{1}{M} \sigma^2 + \frac{M-1}{M} \rho \sigma^2 \quad \left(\dots = \mathbb{E}_{xy} [(f^{[M]}(\mathbf{x}))^2] \right)$$

$$\begin{aligned} \mathbb{E}_{xy} [\Delta (f^{[M]}(\mathbf{x}))] &= \frac{1}{M} \sum_m \mathbb{E}_{xy} \left[(b^{[m]} - f^{[M]}(\mathbf{x}))^2 \right] \\ &= \frac{1}{M} \left(M \mathbb{E}_{xy} [(b^{[m]})^2] + M \mathbb{E}_{xy} [(f^{[M]}(\mathbf{x}))^2] - 2M \mathbb{E}_{xy} [b^{[m]} f^{[M]}(\mathbf{x})] \right) \\ &= \sigma^2 + \mathbb{E}_{xy} [(f^{[M]}(\mathbf{x}))^2] - 2 \frac{1}{M} \sum_{m'} \underbrace{\mathbb{E}_{xy} [b^{[m]} b^{[m']}]}_{=\text{Cov}_{xy} [b^{[m]}, b^{[m']}]} \\ &= \sigma^2 + \left(\frac{1}{M} \sigma^2 + \frac{M-1}{M} \rho \sigma^2 \right) - 2 \left(\frac{M-1}{M} \rho \sigma^2 + \frac{1}{M} \sigma^2 + 0 \cdot 0 \right) \\ &= \frac{M-1}{M} \sigma^2 (1 - \rho) \end{aligned}$$



IMPROVING BAGGING II



$$\mathbb{E}_{xy} \left[L \left(y, f^{[M]}(\mathbf{x}) \right) \right] = \frac{1}{M} \sum_m \mathbb{E}_{xy} \left[L \left(y, b^{[m]} \right) \right] - \mathbb{E}_{xy} \left[\Delta \left(f^{[M]}(\mathbf{x}) \right) \right]$$
$$\mathbb{E}_{xy} \left[\Delta \left(f^{[M]}(\mathbf{x}) \right) \right] \cong \frac{M-1}{M} \text{Var}_{xy} \left[b^{[m]} \right] \left(1 - \text{Corr}_{xy} \left[b^{[m]}, b^{[m]'} \right] \right)$$

- ⇒ **better base learners** are better (... duh)
- ⇒ **more base learners** are better (theoretically, at least...)
- ⇒ **more variable base learners** are better (as long as their risk stays the same, of course!)
- ⇒ **less correlation between base learners** is better:
bagging helps more if base learners are wrong in different ways so that their errors “cancel” each other out.